

AI-Based Crop Disease Prediction for Farmers

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Abstract—Rapid and accurate detection of crop diseases is critical for maintaining agricultural productivity and ensuring food security. Traditional manual inspection methods are prone to human error, while most AI-based models focus on single-crop datasets and lack scalability. In this paper, we propose a deep Convolutional Neural Network (CNN) model for multi-crop disease detection across wheat, rice, and maize. We merged three publicly available datasets with varying class distributions and applied a preprocessing pipeline that included normalization and data augmentation to improve generalization. The model architecture includes multiple convolutional layers, batch normalization, and dropout, trained with a learning rate scheduler to prevent overfitting. Our model achieved a validation accuracy of 84.1% and a final test accuracy of 85.3% on the combined dataset. The resulting system demonstrates potential for real-time disease prediction in field conditions, supporting precision agriculture, and benefiting farmers through early intervention. Future work includes using advanced architectures like Vision Transformers and the integration of explainable AI techniques for transparent decision-making.

Index Terms—Crop Disease Detection, Deep Learning, Convolutional Neural Network, Image Classification, Wheat, Rice, Maize, Agricultural AI

I. INTRODUCTION

Agriculture plays a vital role in global food security and economic development, especially in agricultural countries. Crops are the main source of food and income for millions of farmers, yet they are highly vulnerable to diseases that can devastate yields. It is estimated that up to 40% of crop production is lost annually due to these challenges [1]. Therefore, early detection of crop diseases is important to prevent epidemics, protect yields, and ensure stable food production. Farmers and agriculturalists traditionally rely on manual visual inspection of crops, which is slow and prone to human error. These limitations of conventional methods highlight the need for automated, accurate and real-time disease detection tools that can assist farmers in the earliest stages of infection.

In recent years, advances in artificial intelligence (AI) and computer vision have opened promising opportunities for crop disease monitoring [2]. In particular, deep learning techniques such as CNNs [3] [4] [5] have proven to be remarkably effective at classifying plant diseases from leaf images such as Tomato Yellow Leaf Curl Virus [6], Potato Late Blight [7], and Apple Rust [6] with accuracy exceeding 95%. In controlled experiments, models have achieved over 95%–99% classification accuracy on standard datasets of diseased leaves [8]. Transfer learning with state-of-the-art CNN architectures has achieved impressive results; for example, a fine-tuned EfficientNet-B4 model reached about 94.3% accuracy on a set of plant disease images. Beyond CNNs, transformer-based architectures have recently been introduced to plant disease detection [2].

Despite these advances, the practical adoption of AI for crop disease diagnosis at the farm level faces several challenges. Many existing models are developed on single-crop datasets under lab settings, and their real-time performance in the field remains limited. Farmers require an easy-to-use, real-time diagnostic tool that can work across multiple crops and alert them to disease early, without requiring expert intervention. In this project, we address this need by developing a unified AI-based crop disease classifier. The system is designed to quickly classify common diseases in wheat, rice, and maize.

For this purpose, data from multiple sources is combined. As shown in Figure 1, the dataset includes samples of wheat, rice, and maize leaves. The proposed model is a deep CNN architecture customized for this multi-crop disease classification task. The network consists of several layers to extract discriminative features, with batch normalization and dropout regularization incorporated to improve generalization.

The key contributions of this work include:

- A strategy for merging three distinct plant disease image datasets (wheat, rice, and maize) into one

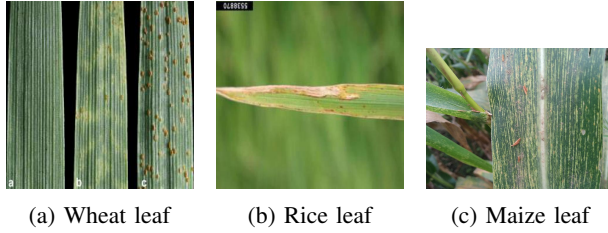


Fig. 1: Sample leaf images from the dataset showing diseased or healthy examples from wheat, rice, and maize.

unified dataset, handling inconsistent splits and class imbalances through augmentation and normalization.

- A custom deep CNN with multiple layers, enhanced with dropout and batch normalization, to effectively learn the characteristics of different crop diseases.
- Emphasis on real-time and in-field usage by targeting common diseases of main crops.

II. LITERATURE REVIEW

This section discusses recent research on the multi-class classification of crop diseases, including common issues such as leaf blight, powdery mildew, and healthy crops.

Paymode et al. [9] applied transfer learning with a VGG16-based CNN. Using a combined dataset of different plants (including grape and tomato leaves), their model achieved high accuracy (98.4% on grape leaf disease and 95.7% on tomato) by leveraging pre-trained features. However, the approach required fine-tuning the model separately for each crop. The current project addresses this limitation by developing a unified model that can handle multiple crops without per-crop retraining.

Xu et al. [10] proposed a novel transfer learning strategy under limited data scenarios. They pre-trained a Vision Transformer on a large plant image collection (PlantCLEF2022) and then fine-tuned it on 12 small plant disease datasets, achieving a mean accuracy of 86.3% across those datasets with only 20 training samples per class. This approach outperformed prior methods by over 12% in similar few-shot settings. However, the method may not scale well to very large or highly diverse datasets due to its specialized tuning. The current project reduces this limitation by using data augmentation and focusing on a selected dataset of key diseases.

Shoaib et al. [8] developed a deep learning method that combines leaf image segmentation and classification. They trained a CNN on a tomato leaf image dataset and reported around 99% accuracy in identifying six tomato leaf diseases after segmenting the diseased regions. Its

key limitation is that it was specialized to tomato leaves and showed performance degradation with noisy or field images. The current project addresses these issues by extending disease prediction to multiple crop types and employing image augmentation and noise-reduction techniques.

Kabala et al. [2] introduced a federated learning approach to preserve farmer data privacy. They trained deep models (CNNs and Vision Transformers) on distributed devices using the PlantVillage dataset and achieved a classification accuracy comparable to centralized training (approximately 98% on common plant disease classes). The limitation is the requirement of high device coordination and computational power.

Shoaib et al. [11] implemented an advanced deep learning model. Using public benchmark datasets (such as an expanded PlantVillage set), their CNN-based system obtained over 95% accuracy in classifying multiple disease categories. The authors noted a limitation in that the model's performance declined with low-quality or noisy images, reflecting challenges with data irregularity. The current project addresses this by including image preprocessing (e.g., filtering and enhancement) and ensuring the model is trained on augmented data.

Shahoveisi et al. [12] combined classical image processing techniques with transfer learning. They evaluated four pre-trained CNN models (Xception, ResNet50, EfficientNetB4, MobileNet) on a dataset of rust-infected vs. healthy leaf images captured in real field and greenhouse conditions. The best model (EfficientNetB4) achieved 94.3% average accuracy in detecting rust disease, aided by preprocessing steps like noise reduction, image resizing, and data augmentation. The approach required careful hyperparameter tuning. The current project addresses this limitation by establishing a more automated and generalized pipeline.

Shafik et al. [13] developed an ensemble-based plant disease detection model using transfer learning. They fine-tuned nine pre-trained CNN architectures (such as ResNet, DenseNet, and EfficientNet) on 15 classes of the PlantVillage dataset and combined them via an adaptive ensemble. The best ensemble (a voting fusion) achieved a high accuracy of about 97.8% in classifying various plant diseases. It introduced increased computational cost and complexity, with a risk of overfitting due to multiple models. The current project addresses this limitation by using a single efficient deep learning model with regularization techniques.

Jafar et al. [14] surveyed various AI techniques for detecting plant diseases in crops like tomato, chili, and potato. The paper explored methods including image preprocessing, feature extraction, and deep learning-based classification, highlighting the benefits of automation in reducing manual effort. Although their review covered

multiple datasets and showed high detection accuracy across crops, the generalized nature of the models discussed may overlook disease-specific details. The current project improves on this by focusing on fine-grained classification across specific diseases in wheat, rice, and maize, ensuring both breadth and diagnostic precision.

Jung et al. [15] designed a three-step deep learning pipeline. Their system uses specialized pre-trained models (e.g., ResNet50 and EfficientNet). This multi-stage model achieved over 97% accuracy in distinguishing healthy leaves and multiple diseases across the three crops. However, the method is highly dataset-dependent and required a large volume of images for training. The current project handles this limitation by using transfer learning and augmentation to reduce data requirements.

Dolatabadian et al. [16] conducted a literature review. They summarized various approaches, from conventional classifiers to deep CNN models, and discussed data preprocessing and feature selection methods used in past research. The authors observed that modern deep learning approaches often produce high accuracy (91.37% on apple leaves) in standard datasets such as PlantVillage. They also pointed out common issues such as class imbalance, overfitting, and lack of model clarity in existing studies. A major limitation of this work is that it is theoretical. The current project addresses this gap by building and evaluating a practical AI-based disease prediction system.

The comparative analysis is presented in Table I which compares the proposed model with existing plant disease detection methods across multiple crops.

III. MATERIALS AND METHODS

This section outlines the materials and methodologies used in this study for the multi-class classification of crop diseases affecting wheat, rice, and maize. Each subsection below details the steps involved in developing and validating the model.

A. Data Preprocessing

We utilized three crop disease image datasets (wheat [17], maize [18], and rice [19]). Each dataset was originally split into training, validation, and test sets (approximately 84%/10%/5% for wheat, 70%/20%/10% for maize, and 87%/8%/4% for rice, respectively). To create a unified dataset for model training, we merged the three datasets, combining their images and labels into a single collection. Each dataset's class labels were provided in separate CSV files; we concatenated these label files and applied one-hot encoding to represent each image's class label in a format suitable for model training. This combination increased the overall variety and number of samples available for learning.

The final unified dataset consisted of the following nine disease classes across all three crops:

TABLE I: Comparison of the proposed CNN-based crop disease classification model with existing state-of-the-art plant disease detection methods across multiple crops

Paper	Dataset	Model	Limitation	Accuracy
Paymode et al. (2021) [9]	Multiple crop datasets	Transfer Learning, Inception, ImageNet	May require fine-tuning per crop	Grapes: 98.40%, Tomatoes: 95.71%
Xu et al. (2022) [10]	Small-scale datasets	Transfer Learning, Fine-tuning	May not scale well to large diverse datasets	Mean: 86.29%
Shoaib et al. (2022) [8]	Tomato Leaf Dataset	CNNs, Image Segmentation	Limited to tomato diseases; Performance drop in noisy images	99.95% (Binary), 99.12% (Six-class)
Kabala et al. (2023) [2]	PlantVillage	Federated Learning, CNNs, ViT	Requires high device coordination and computational power	99.66% (CNN), 96.46% (9-layer CNN)
Shoaib et al. (2023) [11]	PlantVillage	CNNs, Deep Learning	Might struggle with low-quality images	95%
Shahoveisi et al. (2023) [12]	PlantVillage	ResNet_SGD, Transfer Learning	May require parameter tuning for different datasets	EfficientNetB4: 94.29%, ResNet50: 93.52%
Shafik et al. (2023) [13]	PlantVillage	Transfer Learning, Ensemble CNNs	Increased computational cost; Possible overfitting	99.75%
Jafar et al. (2023) [14]	Multiple crop leaf datasets	AI-based detection, Image Classification	Generalized model may lack specific disease details	87% (CNN on PlantVillage)
Jung et al. (2023) [15]	Leaves of Bell Pepper, Potato, Tomato	ResNet50, EfficientNet, Deep Learning	Dataset-dependent; May require large training data	97.09%
Dolatabadian et al. (2024) [16]	Reviewed datasets incl. PlantVillage	Literature Review, CapsNet	No implementation; theoretical overview	91.37% (on apple leaves)
Proposed Work	Combined datasets of wheat, rice, and maize leaves (merged from three public sources)	CNN with data augmentation, dropout, batch normalization, learning rate scheduling	-	Test: 85.3%, Validation: 84.1%

- Wheat: Leaf Blight, Leaf Rust, Healthy
- Rice: Bacterial Leaf Blight, Brown Spot, Healthy
- Maize: Cercospora Leaf Spot, Common Rust, Healthy

All images experienced preprocessing before being fed into the model. Pixel intensity values were normalized (scaled to [0,1]) to standardize the input range. We also applied various data augmentation techniques (commonly used in plant disease imaging to improve generalization [20]) to increase data variability, including:

- **Random rotations:** rotating images at various angles to imitate different leaf orientations.
- **Flipping:** horizontal and vertical flips to account for leaf symmetry and viewpoint differences.
- **Shifting:** small translations of the image (horizontal/vertical shifts) to make the model strong to leaf position changes.
- **Zooming:** random zoom-in and zoom-out transformations to copy varying camera distances or image scales.

After augmentation, all images were resized to 224x224 pixels to meet the input size requirements of the CNN-based model. These preprocessing steps ensured consistent image dimensions and enhanced the model's ability to generalize across different conditions.

B. Methodology

We designed a deep CNN [3] [4] [5] for crop disease classification, as CNN models have displayed high accuracy in plant disease recognition tasks [20]. The model was implemented as a custom sequential architecture consisting of four convolutional blocks. Each block comprised a Conv2D layer with a 3×3 kernel (using ReLU activation) followed by a batch normalization layer, a max-pooling layer for spatial downsampling, and a dropout layer for regularization. With each successive block, the number of convolution filters was increased to capture more complex features, while the pooling layers progressively reduced the angular resolution of feature maps. After the four convolutional blocks, the feature maps were flattened and passed through fully connected dense layers. Finally, a softmax output layer produced a probability distribution over the disease classes for prediction. The architecture of the CNN model is illustrated in Figure 2.

The model was trained using the Adam optimizer (with an initial learning rate of 0.001) and optimized with categorical cross-entropy loss appropriate for multi-class classification. An adaptive learning rate scheduler was employed to adjust the learning rate during training: the learning rate was automatically reduced when the validation loss stabilized, which helped improve convergence stability. Model training proceeded by iteratively updating weights via back-propagation to minimize the cross-entropy loss. We monitored validation performance and saved the best-performing model for evaluation.

As the input images propagate through the CNN, the network learns to extract features in a hierarchical manner. Early convolutional layers capture low-level visual patterns such as edges and color textures. In contrast, deeper layers learn higher-level, abstract representations of the leaf images that correspond to disease-specific patterns [21]. Through successive convolution and pooling operations, the model focuses on the most informative regions of the image. By the final convolutional layers, distinctive disease lesions on leaves (such as localized leaf spots, regions of discoloration, or fungal-infected areas) are effectively recognized and emphasized in the feature maps. We observed that the model’s predictions often relied on these high-contrast lesion features, especially when they appeared prominently in central or mid-leaf regions of the image. This behavior aligns with domain knowledge and prior findings, indicating that the CNN learned to prioritize important visual symptoms of crop diseases in a manner similar to expert human assessment [21].

To summarize the training pipeline, Algorithm 1 describes the key steps involved in developing the model.

Algorithm 1 Crop Disease Detection via CNN-based Deep Learning

- 1: **Input:** Labeled leaf image dataset
 $D = \{(x_i, y_i)\}_{i=1}^n$
 - 2: **Output:** Trained CNN model for disease classification
 - 3: Preprocess images: resize to 224×224, normalize pixel values
 - 4: Augment images: apply random rotation, flip, shift, and zoom
 - 5: Initialize CNN model with defined architecture
 - 6: **for** each training epoch **do**
 - 7: Update learning rate based on the step-decay scheduler
 - 8: Train model using categorical cross-entropy on the training set
 - 9: Evaluate on the validation set
 - 10: **if** validation accuracy improves **then**
 - 11: Save model checkpoint
 - 12: **end if**
 - 13: **end for**
 - 14: Evaluate the final model on the test set and compute the confusion matrix
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IV. RESULTS AND DISCUSSION

All experiments and model training were conducted on an Apple Mac mini equipped with the Apple M2 Pro chip. This hardware features a 12-core CPU (8 performance cores + 4 efficiency cores) and a 19-core integrated Apple GPU with Metal 3 support. The system has 32 GB of RAM and runs macOS with GPU acceleration enabled via Apple’s Metal framework. Training was performed on the GPU to take advantage of hardware acceleration, ensuring faster computation during the model’s training phase.

To assess the classification performance, we evaluated standard metrics including accuracy, precision, recall, and F1-score.

- **Accuracy:** The ratio of correctly predicted instances to the total instances. It measures the overall correctness of the model and is given by:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision:** The proportion of predicted positive cases that are positive. It indicates the accuracy of positive predictions and is defined as:

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall:** The proportion of actual positive cases (diseased instances) that are correctly identified by

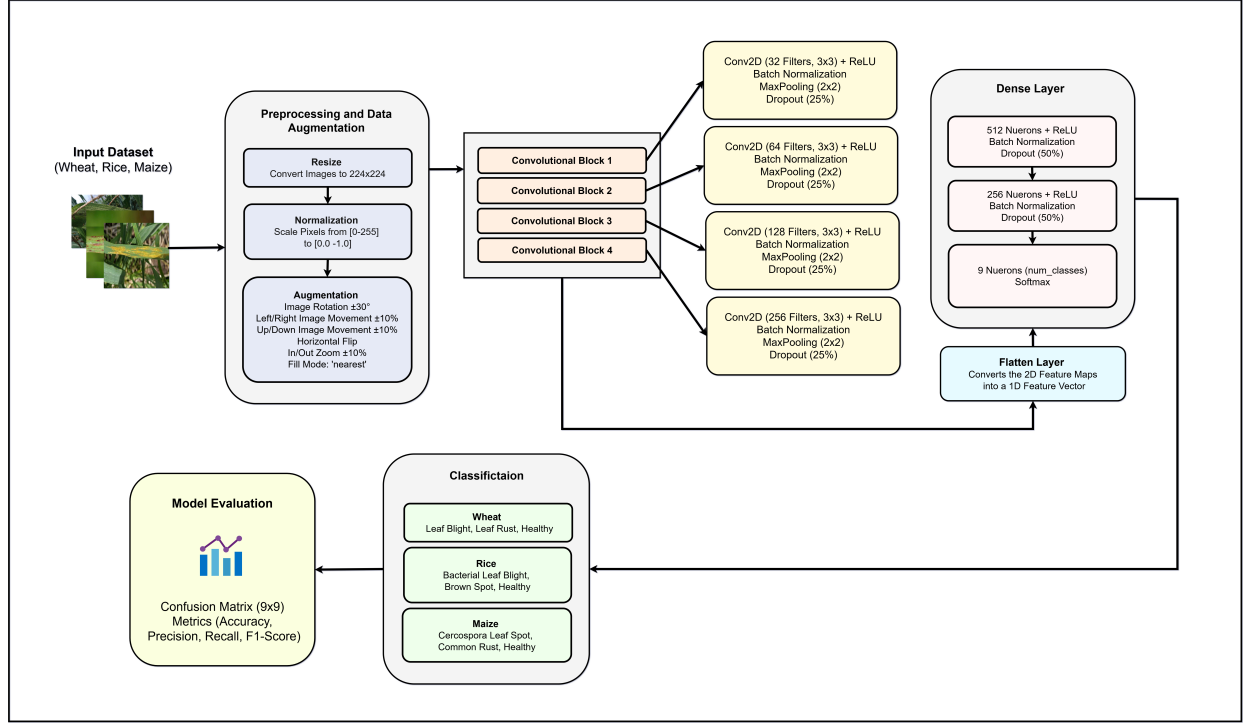


Fig. 2: Proposed system architecture consists of image input preprocessing, feature extraction via convolutional blocks, classification using fully connected layers, and prediction through softmax activation.

the model. It measures the model's ability to find all relevant cases and is computed as follows:

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1-Score:** The harmonic mean of precision and recall, offering a balance between the two. It provides a single score that weights both false positives and false negatives:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The confusion matrix provides a summary of the model's performance on the test dataset, illustrating both correct and incorrect predictions. Figure 3 presents the matrix with actual and predicted labels.

The confusion matrix shows the classification performance. The highest classification accuracies were observed for MLB (487/500) and MSV (495/500), indicating the model's strong ability to distinguish maize-related diseases. Similarly, high accuracy was achieved for Brownspot (144/150) and Blight (142/150), confirming effective recognition of rice leaf symptoms. Moderate performance was noted for Leaf Rust and Blast, though some misclassifications occurred between visually similar diseases. Errors were most common between Barley Yellow Dwarf, Healthy Wheat, and MSV, where overlapping symptoms (e.g., leaf discoloration)

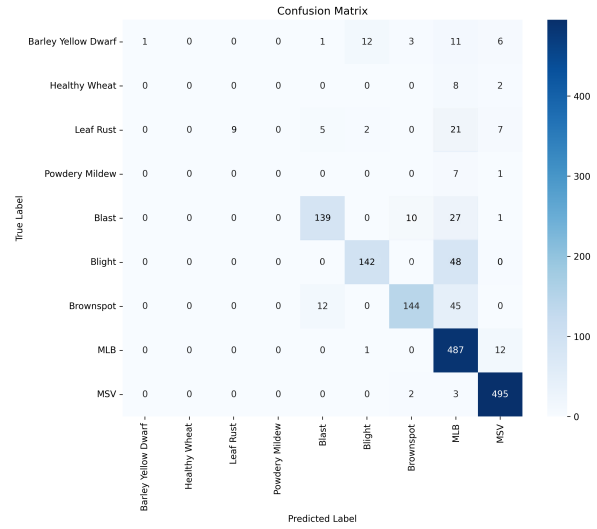
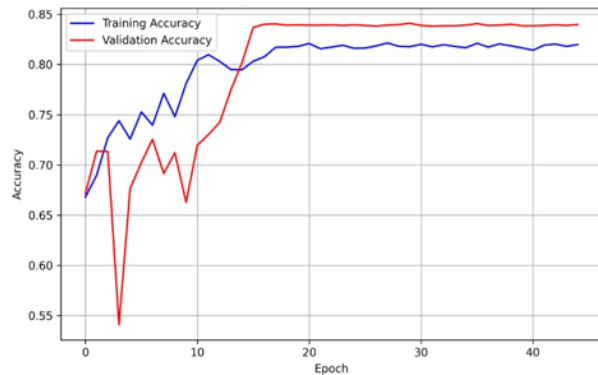


Fig. 3: Confusion Matrix showing the performance of the proposed model for the classification of Barley Yellow Dwarf, Healthy Wheat, Leaf Rust, Powdery Mildew, Blast, Blight, Brownspot, MLB, and MSV categories.

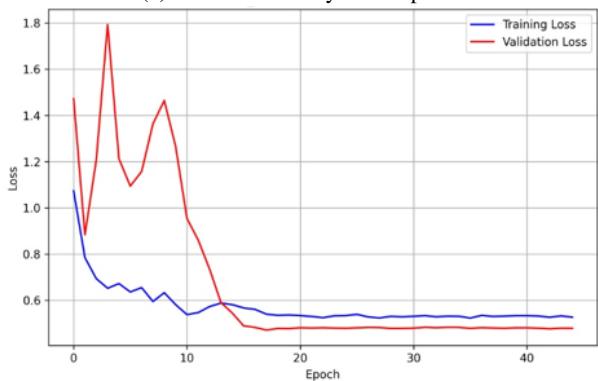
may have contributed to confusion. However, the diagonal dominance of the matrix confirms that the model maintained high sensitivity and precision for the majority of disease classes, validating its generalization across

wheat, rice, and maize crops.

Key hyperparameters were tuned to optimize performance. We used a batch size of 32, which provided a good balance. The learning rate was initially set to 0.001 (with the Adam optimizer). This value allowed the model to learn steadily without diverging. Additionally, an early stopping criterion was used. We also experimented with different optimizer settings (comparing Adam vs. SGD momentum) and found Adam with default momentum parameters gave faster convergence for this problem.



(a) Model Accuracy over Epochs



(b) Model Loss over Epochs

Fig. 4: Training and validation accuracy/loss curves over 45 epochs showing model convergence and performance stability.

We tracked the model’s learning progress over successive epochs using training and validation accuracy/loss curves. As illustrated in Figure 4a, the model’s training and validation accuracy improve rapidly in the initial epochs. The validation accuracy (red curve) climbs steeply and peaks around 85% at approximately the 15th epoch. This indicates that the model learns the most important features early, with validation accuracy converging to the training accuracy (blue curve) and remaining steady thereafter. Such behavior suggests that the model’s capacity is well-matched to the task, as the learning does not stop too early (which would suggest

underfitting) and the validation accuracy sustains close to the training accuracy without significant divergence (suggesting minimal overfitting). In Figure 4b, the loss curves show a corresponding trend. The training loss (blue) frequently declines, while the validation loss (red) shows a sharp drop around epochs 10 to 12, reaching a low point by about the 12th epoch. After this point, the validation loss remains low and nearly parallel to the training loss.

The results indicate that our CNN-based approach effectively learned discriminative features of crop diseases, achieving an overall test accuracy of 85.28%. The alignment of training and validation curves suggests that the model did not face severe overfitting. In the final epochs, training accuracy was slightly higher than validation accuracy which is expected as the model fits the training data. This slight gap between training and validation performance is a sign of minor overfitting in later stages; however, it was reduced by our use of dropout regularization and early stopping. The absence of a large divergence between training and validation accuracy curves suggests that underfitting was not a major issue. The chosen number of epochs (with early stopping) ensured we stopped training at an optimal point before the model started overfitting considerably.

We compared our CNN model’s performance with simpler baseline approaches to measure the advantages of deep learning [22] for this task. A conventional machine learning baseline (for example, an SVM classifier using hand-crafted color and texture features of leaves [23]) was found to be far less effective, with accuracy levels in the range of only 60–70% in initial trials. A smaller CNN with only a few layers trained on the same data struggled to exceed 70–75% accuracy [24], as it lacked the depth to capture the variability of disease appearances. In contrast, our deeper CNN architecture [3] [4] [5], with appropriate regularization and tuning, achieved 85%+ accuracy by effectively learning both low-level leaf textures and higher-level disease patterns. As shown in Figure 5, different crop images were passed through the model.

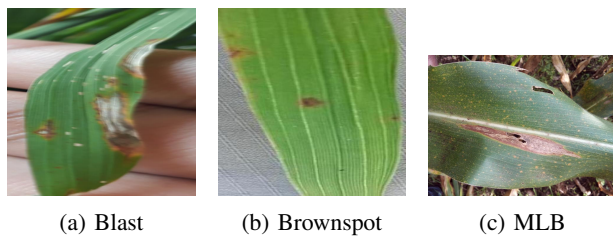


Fig. 5: Examples of input crop images and their predicted classes as labeled by the model.

The decision to use a CNN [3] [4] [5] was driven by

the nature of the task. Plant disease recognition relies on identifying patterns like spots, discolorations, or lesions on leaves; a CNN's layered structure (convolutions and pooling) is well-suited to detect these features at multiple scales. In our implementation, we selected a CNN architecture with moderate depth and complexity appropriate for the dataset size.

V. CONCLUSION

This study presents a CNN-based model for automated crop disease classification using merged datasets of wheat, rice, and maize. The model achieved a test accuracy of 85.28%, demonstrating strong generalization across all crops with minimal overfitting. By addressing the limitations of manual disease detection, this work supports the need for AI-driven tools in precision agriculture. Future directions include testing advanced architectures like Vision Transformers [25], expanding datasets to new crops and conditions, and combining explainable AI for transparent decision support.

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